Question ID: 8de51658

Question

Text 1

The idea that time moves in only one direction is instinctively understood, yet it puzzles physicists. According to the second law of thermodynamics, at a macroscopic level some processes of heat transfer are irreversible due to the production of entropy—after a transfer we cannot rewind time and place molecules back exactly where they were before, just as we cannot unbreak dropped eggs. But laws of physics at a microscopic or quantum level hold that those processes *should* be reversible.

Text 2

In 2015, physicists Tiago Batalhão et al. performed an experiment in which they confirmed the irreversibility of thermodynamic processes at a quantum level, producing entropy by applying a rapidly oscillating magnetic field to a system of carbon-13 atoms in liquid chloroform. But the experiment “does not pinpoint ... what causes [irreversibility] at the microscopic level,” coauthor Mauro Paternostro said.

Based on the texts, what would the author of Text 1 most likely say about the experiment described in Text 2?

Answer

A. It would suggest an interesting direction for future research were it not the case that two of the physicists who conducted the experiment disagree on the significance of its findings.

B. It provides empirical evidence that the current understanding of an aspect of physics at a microscopic level must be incomplete.

C. It is consistent with the current understanding of physics at a microscopic level but not at a macroscopic level.

D. It supports a claim about an isolated system of atoms in a laboratory, but that claim should not be extrapolated to a general claim about the universe.

Question ID: 82c05b34

Question

Text 1

The live music festival business is growing in event size and genre variety. With so many consumer options, organizers are finding ways to cement festival attendance as a special experience worth sharing. This phenomenon is linked to the growing “experiential economy,” where many find it gratifying to purchase lived experiences. To ensure a profitable event, venues need to consider the overall consumer experience, not just the band lineup.

Text 2

Music festival appearances are becoming a more important part of musicians’ careers. One factor in this shift is the rising use of streaming services that allow access to huge numbers of songs for a monthly fee, subsequently reducing sales of full-length albums. With this shift in consumer behavior, musicians are increasingly dependent on revenue from live performances.

Based on the texts, both authors would most likely agree with which statement?

Answer

A. Consumers are more interested in paying subscription fees to stream music than in attending music festivals in person.

B. Consumers’ growing interest in purchasing experiences is mostly confined to the music industry.

C. Changing consumer behaviors are leading to changes in music-related businesses.

D. The rising consumer demand for live music festivals also generates higher demand for music streaming platforms.

Question ID: f1c9d2c1

Question

Text 1

Stage lighting theorist Adolphe Appia was perhaps the first to argue that light must be considered alongside all the various elements of a stage to create a single, unified performance. Researcher Kelly Bremner, however, has noted that Appia lacked technical expertise in the use of light in the theater. As a result of Appia's inexperience, Bremner argues, Appia's theory of light called for lighting practices that weren't possible until after the advent of electricity around 1881.

Text 2

Adolphe Appia was not an amateur in the practice of lighting. Instead, it is precisely his exposure to lighting techniques at the time that contributed to his theory on the importance of light. When working as an apprentice for a lighting specialist in his youth, Appia observed the use of portable lighting devices that could be operated by hand. This experience developed his understanding of what was possible in the coordination of elements on the stage.

Based on the texts, how would the author of Text 2 most likely respond to the claim about Appia's level of technical expertise made by Bremner in Text 1?

Answer

A. Many lighting technicians dismissed Appia's ideas about light on the stage.

B. Appia likely gained a level of technical expertise during his time as an apprentice.

C. Theater practitioners who worked with Appia greatly admired his work.

D. Appia was unfamiliar with the use of music and sound in theater.

Question ID: eae66bf9

Question

Text 1

In 2021, a team led by Amir Siraj hypothesized that the Chicxulub impactor—the object that struck the Yucatán Peninsula sixty-six million years ago, precipitating the mass extinction of the dinosaurs—was likely a member of the class of long-period comets. As evidence, Siraj cited the carbonaceous chondritic composition of samples from the Chicxulub impact crater as well as of samples obtained from long-period comet Wild 2 in 2006.

Text 2

Although long-period comets contain carbonaceous chondrites, asteroids are similarly rich in these materials. Furthermore, some asteroids are rich in iridium, as Natalia Artemieva points out, whereas long-period comets are not. Given the prevalence of iridium at the crater and, more broadly, in geological layers deposited worldwide following the impact, Artemieva argues that an asteroid is a more plausible candidate for the Chicxulub impactor.

Based on the texts, how would Artemieva likely respond to Siraj's hypothesis, as presented in Text 1?

Answer

A. By insisting that it overestimates how representative Wild 2 is of long-period comets as a class

B. By arguing that it does not account for the amount of iridium found in geological layers dating to the Chicxulub impact

C. By praising it for connecting the composition of Chicxulub crater samples to the composition of certain asteroids

D. By concurring that carbonaceous chondrites are prevalent in soil samples from sites distant from the Chicxulub crater

Question ID: 9a94eb77

Question

Text 1

One challenge faced by researchers studying global urbanization is that countries may define urban settlements differently. Many countries define urban settlements based on the number of people living in them. However, countries do not all use the same number; for example, Iceland uses a minimum population of 200, while the Netherlands uses 20,000. Other countries use a combination of population and other factors. This variation makes it difficult for researchers to compare aspects of urbanization in different countries.

Text 2

Recently, a group of six international organizations developed global definitions of common types of settlements. The group developed a new measure called the “degree of urbanization.” This new measure establishes global criteria used to define three types of settlements (cities, towns, and rural areas) and allows researchers to better understand global urbanization rates.

Based on the texts, how would the author of Text 2 most likely respond to the problem presented in Text 1?

Answer

A. By recommending that a specific institution should further investigate the problem

B. By suggesting that researchers focus on topics besides urbanization

C. By supplying additional ways in which urbanization research is difficult

D. By noting that a possible solution to the problem is available

Question ID: 03080769

Question

Text 1

Philosopher G.E. Moore's most influential work entails the concept of common sense. He asserts that there are certain beliefs that all people, including philosophers, know instinctively to be true, whether or not they profess otherwise: among them, that they have bodies, or that they exist in a world with other objects that have three dimensions. Moore's careful work on common sense may seem obvious but was in fact groundbreaking.

Text 2

External world skepticism is a philosophical stance supposing that we cannot be sure of the existence of anything outside our own minds. During a lecture, G.E. Moore once offered a proof refuting this stance by holding out his hands and saying, "Here is one hand, and here is another." Many philosophers reflexively reject this proof (Annalisa Coliva called it "an obviously annoying failure") but have found it a challenge to articulate exactly why the proof fails.

Based on the texts, how would the author of Text 1 most likely respond to proponents of the philosophical stance outlined in Text 2?

Answer

A. By agreeing with those proponents that Moore's treatment of positions that contradict his own is fundamentally unserious

B. By suggesting that an instinctive distaste for Moore's position is preventing external world skeptics from constructing a sufficiently rigorous refutation of Moore

C. By arguing that if it is valid to assert that some facts are true based on instinct, it is also valid to assert that some proofs are inadequate based on instinct

D. By pointing out that Moore would assert that external world skepticism is at odds with other beliefs those proponents must unavoidably hold

Question ID: 12d81fc1

Question

Text 1

Because literacy in Nahuatl script, the writing system of the Aztec Empire, was lost after Spain invaded central Mexico in the 1500s, it is unclear exactly how meaning was encoded in the script's symbols. Although many scholars had assumed that the symbols signified entire words, linguist Alfonso Lacadena theorized in 2008 that they signified units of language smaller than words: individual syllables.

Text 2

The growing consensus among scholars of Nahuatl script is that many of its symbols could signify either words or syllables, depending on syntax and content at any given site within a text. For example, the symbol signifying the word *huipil* (blouse) in some contexts could signify the syllable "pil" in others, as in the place name "Chipiltepec." Thus, for the Aztecs, reading required a determination of how such symbols functioned each time they appeared in a text.

Based on the texts, how would the author of Text 2 most likely characterize Lacadena's theory, as described in Text 1?

Answer

A. By praising the theory for recognizing that the script's symbols could represent entire words

B. By arguing that the theory is overly influenced by the work of earlier scholars

C. By approving of the theory's emphasis on how the script changed over time

D. By cautioning that the theory overlooks certain important aspects of how the script functioned

Question ID: 27d9bb69

Question

Text 1

Many studies in psychology have shown that people seek out information even when they know in advance that they have no immediate use for it and that they won't directly benefit from it. Such findings support the consensus view among researchers of curiosity: namely, that curiosity is not instrumental but instead represents a drive to acquire information for its own sake.

Text 2

While acknowledging that acquiring information is a powerful motivator, Rachit Dubey and colleagues ran an experiment to test whether emphasizing the usefulness of scientific information could increase curiosity about it. They found that when research involving rats and fruit flies was presented as having medical applications for humans, participants expressed greater interest in learning about it than when the research was not presented as useful.

Based on the texts, how would Dubey and colleagues (Text 2) most likely respond to the consensus view discussed in Text 1?

Answer

A. By suggesting that curiosity may not be exclusively motivated by the desire to merely acquire information

B. By conceding that people may seek out information that serves no immediate purpose only because they think they can use it later

C. By pointing out that it is challenging to determine when information-seeking serves no goal beyond acquiring information

D. By disputing the idea that curiosity can help explain apparently purposeless information-seeking behaviors

Question ID: f52cc78c

Question

Text 1

Polar bears sustain themselves primarily by hunting seals on the Arctic sea ice, but rising ocean temperatures are causing the ice to diminish, raising concerns about polar bear population declines as these large predators' seal-hunting habitats continue to shrink. A 2020 study examining polar bear populations across the Arctic concluded that populations affected by sea-ice loss are at great risk of extinction by the end of the twenty-first century.

Text 2

Monitoring carried out by researchers from the Norwegian Polar Institute shows that the polar bear population on the Arctic archipelago of Svalbard remains stable and well nourished despite rapidly declining sea ice in recent years. The researchers attribute this population's resilience in part to a shift in feeding strategies: in addition to hunting seals, the Svalbard polar bears have begun relying on a diet of reindeer meat and birds' eggs.

Based on the texts, how would the researchers in Text 2 most likely respond to the conclusion presented in the underlined portion of Text 1?

Answer

A. By noting that it neglects the possibility of some polar bear populations adapting to changes in their environment

B. By suggesting that it is likely incorrect about the rates at which warming ocean temperatures have caused sea ice to melt in the Arctic

C. By asserting that it overlooks polar bear populations that have not yet been affected by loss of seal-hunting habitats

D. By arguing that it fails to account for polar bears' reliance on a single seal-hunting strategy